



Tracker 2026 Data Taking Anomaly

Faser General Meeting

Pawan

on behalf of Tracker Team

University of Liverpool

11 September 2026



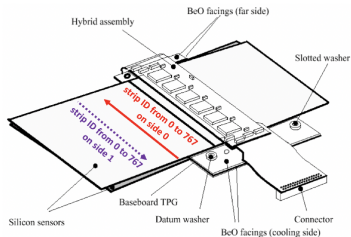
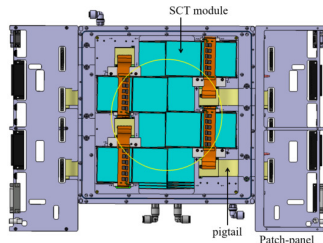
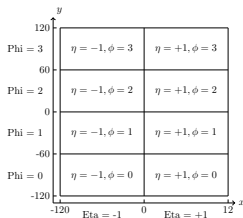
Particle Physics

Introduction

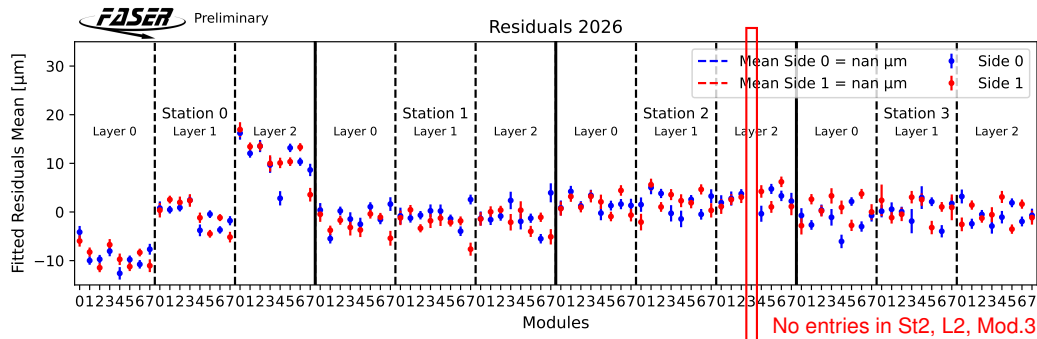
- ❖ The Tracker has been taking data since 2022 and has been doing it quite well.
- ❖ We see some potential issues in the 2026 data taking specifically in a specific module.
- ❖ This presentation will discuss this issue.
- ❖ We are quite early in understanding this.
- ❖ But thought it was best to make everyone aware of the situation.
- ❖ We will discuss the issue in a more “chronological” perspective rather than a diagnostic one.

Quick Recap on Tracker

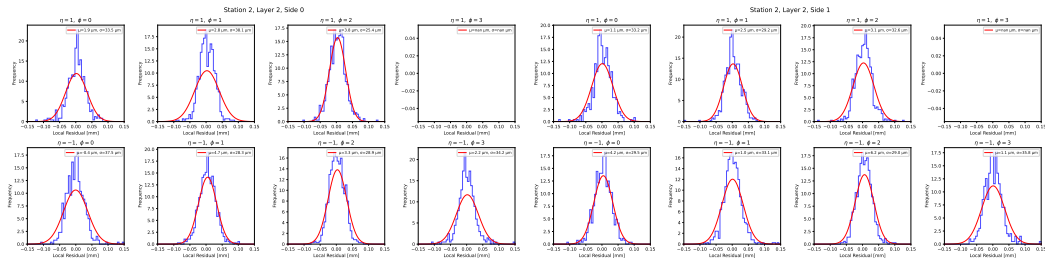
- We have four tracking stations
- Each station has three layers
- Each layer has eight modules
- Each module has two pairs of single-sided silicon micro-strip sensors



- ❖ First seen by Haruhi when trying to calculate unbiased residuals for 2026
- ❖ See Tracker Meeting Aug 12, 2026.

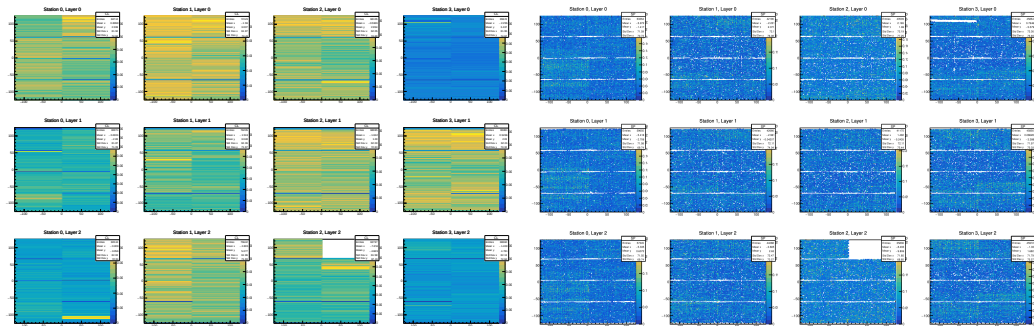


- Initially suspected to be a fit failure/stats issue.



- Quickly ruled out as nearby modules (in same station) had entries.

Looking at Clusters/Space Points



(a) Clusters from Run #022300

(b) Space Points from Run #022300

Figure: SCT Clusters and Space Points from Run #022300. Note that these are subset of the space points that are from the same layer.

Similar Story with SCT Hits itself

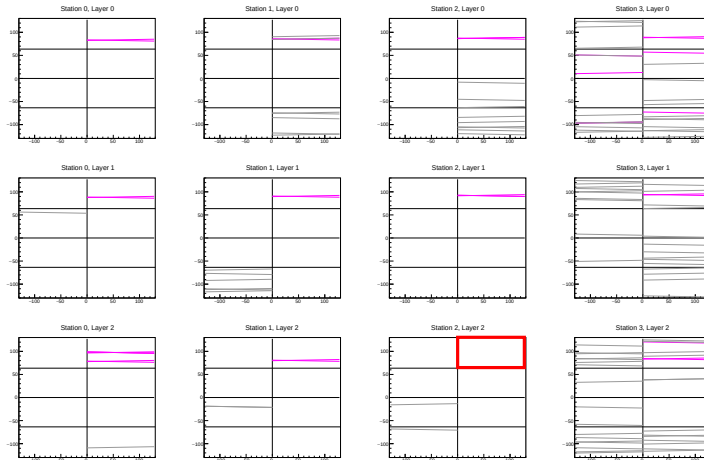


Figure: SCT Hits from Event #2840829, Run #022300. Hits missing in the highlighted module.
Note that out of time HITS (!01X) are shown in grey.

What does the Track look like?

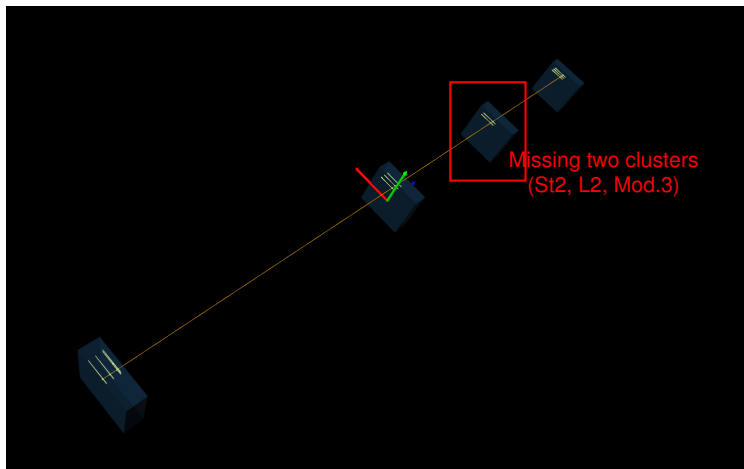
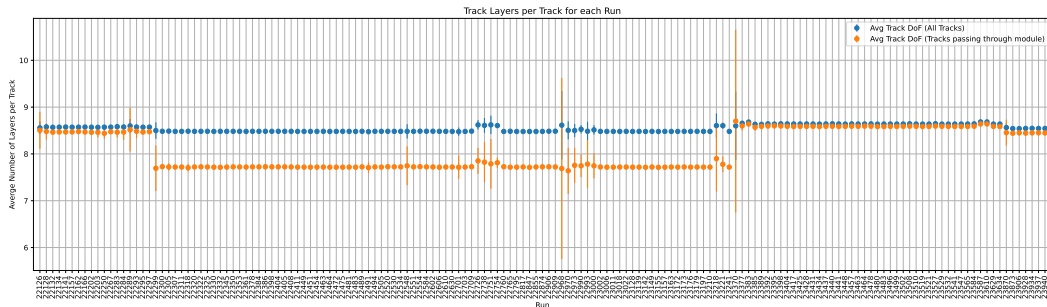


Figure: Event #2840829, Run #022300

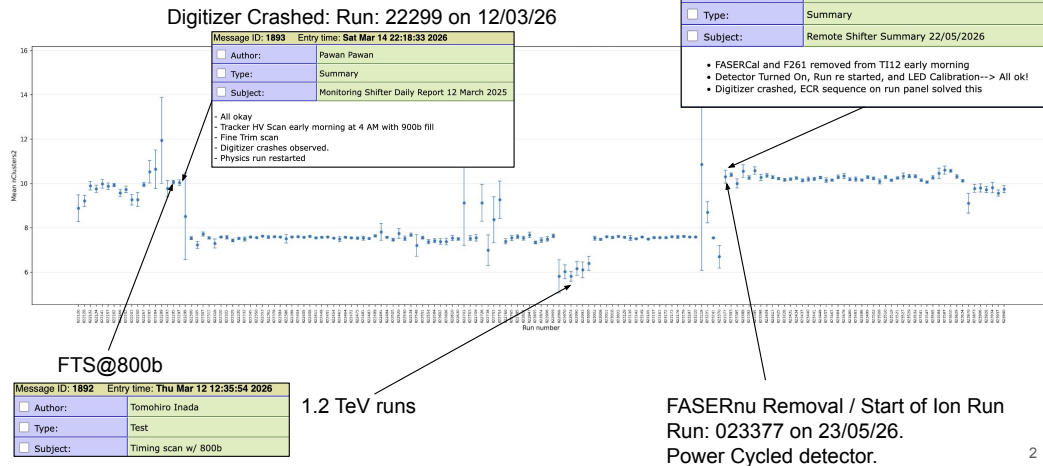
Track Layers



- ❖ Average Layers per Track – Ideally should be 9.
- ❖ Harder to see when looking at all tracks
- ❖ The deficit of one layer more clearer when you require
 - ❖ Track passing through the module in question.
- ❖ Can be seen in the clusters, Track nDoF as well

What could it mean for the Tracking?

- ❖ It appears as though one module does not have any data
 - ❖ This is probably not completely true
- ❖ In theory this should just be a reduction of two measurements in the final track
 - ❖ The Track- χ^2 is lower for these events (Add chi2 plots to backup)
 - ❖ But overall quality (χ^2/NDF) is still good.
 - ❖ Overlap on fiducial volume is around 5.5% (See slides from Oscar in backup)
- ❖ More importantly, does the missing hits cause issues with track reconstruction?
 - ❖ This is a harder question to answer.
 - ❖ Needs a more detailed study to quantify this.
 - ❖ Ideally, shouldn't since the seeding should still pick up everything needed for the track.



Tracker Meeting

Figure: Timeline of events from Oscar see Tracker Meeting Aug 25 2026

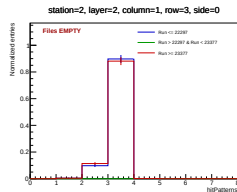
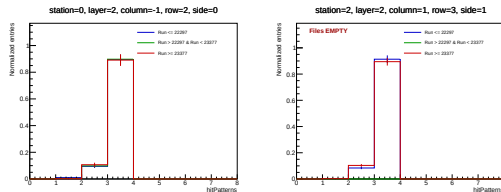
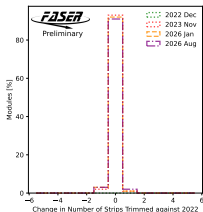
Timeline Summarized

- ❖ Thursday Mar 12 2025: Fine Time Scan, after HV Scan
- ❖ Thursday Mar 12 2025: Fine time adjustment
- ❖ March 13-14 2025: Digitizer Crashes being observed (and restarted)
- ❖ Since then, module seems to have the issue
- ❖ Recovered by Run #023377, at this point we are in heavy-ion runs
- ❖ Roughly corresponds to $XXx \text{ fb}^{-1}$

Tangent: Possible Timing Issues?

[Oscar]

- ❖ Since we adjusted the timing just prior to this
- ❖ We considered the possibility that the data just went into the wrong time bin
- ❖ Oscar quickly ruled this out since we didn't see any data at all.
- ❖ The timing scan itself did not show any abnormalities.



Caveat: How do we read the raw data?

- ❖ We saw that the SCT Raw Data was broken
- ❖ All of the checks so far uses Calypso
 - ❖ Brain pointed out that this implicitly assumes that Calypso decoded the data correctly.
- ❖ This opens up new questions
 - ❖ Is the raw data actually broken? If so, why?
 - ❖ If not, why is Calypso not decoding it correctly?
 - ❖ Why isn't Calypso flagging an error?

Reconstruction

❖ Reconstruction does flag an error...But

ToolSvc.TrackerDataDecoderTool

ERROR Online module 0 reports one or more errors.

❖ This specific error is flagged from here TrackerDataDecoderTool.cxx#L174

```
else if (sctEvent->HasError())
{
    // FIXME: Should we return failure here?
    ATH_MSG_ERROR("Online module " << onlineModuleID << " reports one or more errors.");
    continue;
}
```

- ❖ While we report the error, we don't handle it in anyway, we “continue” over it.
- ❖ This is partly why we didn't catch it earlier in the prompt reconstruction.
 - ❖ this particular module has an error flag set in the raw data.
 - ❖ we don't consider this a failure and we don't load the data for the module.
 - ❖ other modules in the same event are still loaded and processed.
 - ❖ and we still continue the reconstruction albeit with “missing” data.

End Result

- ❖ We have a valid “event” once Calypso is done reading the raw data.
 - ❖ This event does not really have any data from the specific module.
 - ❖ If the track did not pass through the module, then it is not a problem.
 - ❖ But if it did, then we don't use the specific SCT HITS.
- ❖ Only way to stumble on this would be
 - ❖ Either by looking for ERRORS in the log
 - Note that this isn't guaranteed due to messaging limits (see `MessageSvc.cpp#L333`)
 - Do need to check if this is the case here.
 - ❖ Or by specifically looking for the tracks that pass through this module
 - this is not something we usually do
 - we briefly did this for 2024 data (see Track DQ 2024)
 - ❖ Maybe some more data monitoring could help?
- ❖ Ideally if we have an error we should stop and handle it

So... Whats wrong with the raw data?

[Brain]

- ❖ The raw data dump has no errors nominally
 - ❖ Unless you run in debug mode

```
[ERROR] (file = TrackerDataFragment.icc)(func = DecodeModuleData)(line = 271)
| TrackerDataFragment::DecodeModuleData :: Module Data: ERROR code 0x1 for chip 9
```

- ❖ This is the decoding error that was flagged in the reconstruction.
- ❖ Still investigating as to what is the exact cause of this error.

000	Leader
aaaa	Chip address
eee	Error code
	001: No data available (The chip has not received an L1 command)
	010: Buffer Overflow
	100: Buffer Error (Soft Reset is needed.)
1	Separator bit

Table 14: Error data packet: 11 bits.

Figure: Tracker Readout Spec

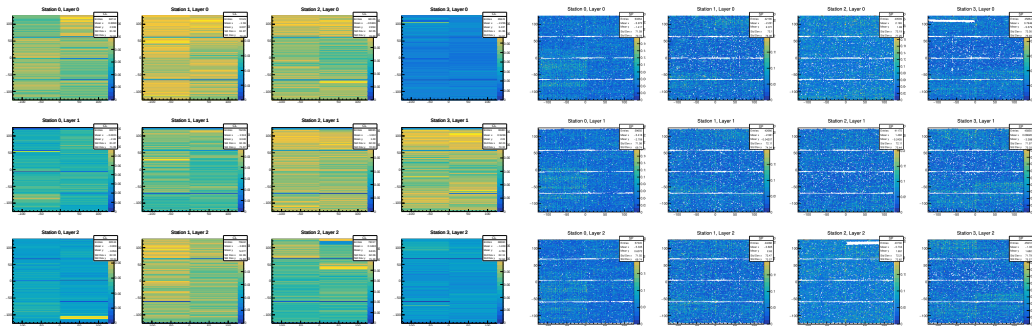
Is it recoverable?

- ❖ It is still unclear.
- ❖ A preliminary test was to comment out this error check in Calypso.

```
else if (sctEvent->HasError())
{
    // FIXME: Should we return failure here?
    ATH_MSG_ERROR("Online module " << onlineModuleID << " reports one or more errors.");
    // continue;
    //Let's fill the data and see what happens
}
```

- ❖ This seems to run **none of the other checks are triggered (? Pawan: Verify this)**
 - ❖ This is probably a good sign.
- ❖ This is not a proper fix, but just a test to see if data is recovered.
 - ❖ Still need to identify and fix the root cause of the error.
 - ❖ If not fixable, then we need to handle it correctly in the reconstruction.
- ❖ Some results in the next slides.

Cluster and Space Points after disabling the check



(a) Clusters from Run #022300

(b) Space Points from Run #022300

Figure: SCT Clusters and Space Points from Run #022300 after disabling the error check mentioned above.

SCT Hits

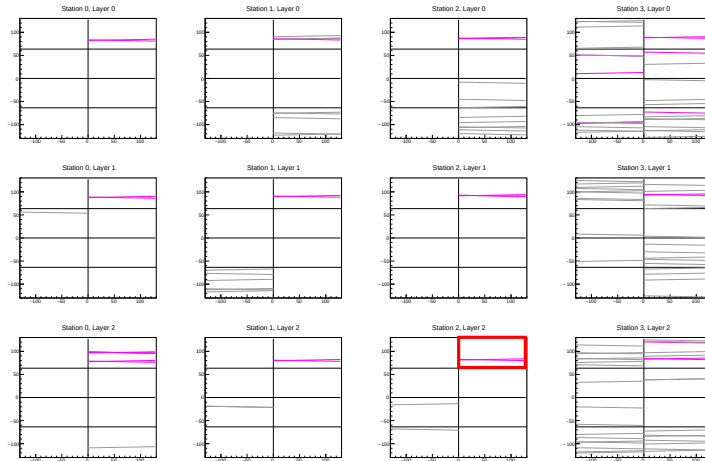


Figure: SCT Hits from Event #2840829, Run #022300 after disabling the error check mentioned above. Hits are restored(?) in the highlighted module.

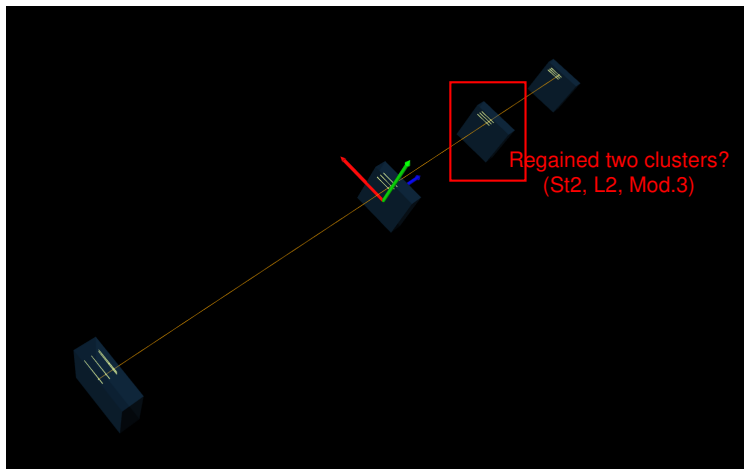


Figure: Event #2840829, Run #022300 after disabling the error check mentioned above.

Closer Look at the Strips

- ❖ So the SCT Hits were recovered for the above event?
- ❖ Is it really recovered everywhere?
- ❖ Pick a run (#022300) and count all the strips that were hit

Nominal

- ❖ 1662 Strips have no hits
- ❖ Of which 1536(?) are in St2, L2, Mod.3
- ❖ Still understanding the numbers here.
- ❖ Module seems to have no data?

With Error Check Disabled

- ❖ 254 Strips have no hits.
- ❖ Of which 128(?) are in St2, L2, Mod.3
- ❖ Strips 512 to 639 have no hits (side 1)
- ❖ Consistent faulty chip? (chip 9?)

- ❖ Possible chip issue was being treated as a module issue by Calypso.
- ❖ Needs some more careful checks.
- ❖ But cautiously optimistic.

Summary

- ❖ Some issues was observed in the 2026 Tracker data.
 - ❖ Specifically in St2, L2, Mod.3 didn't seem to have any hits/clusters
 - ❖ This could probably be isolated to a specific chip
 - ❖ Studies still ongoing to better understand this.
- ❖ The full extent of the impact on tracking is unclear
 - ❖ Probably not too huge
 - ❖ Could use some studies to quantify this.
- ❖ We could use some better error handling in the reco. side
- ❖ Maybe ideas for additional data monitoring/ DQ checks for future
- ❖ Partly recovered by disabling some checks.
 - ❖ This is good which means we can recover at least some of the data if not all.

Impact in FV

1: Circular segment above chord $y = d$

For a chord at distance d from the center, the segment area is

$$A_{\text{seg}}(d) = r^2 \cos^{-1}\left(\frac{d}{r}\right) - d\sqrt{r^2 - d^2} \quad (1)$$

With $d = 60$, $r = 90$:

$$A_{\text{seg}} = (90)^2 \cos^{-1}\left(\frac{60}{90}\right) - 60\sqrt{90^2 - 60^2} \quad (2)$$

$$= 8100 \times 0.8411 - 60 \times 67.08 \quad (3)$$

$$\approx 6813 - 4025 \quad (4)$$

$$\approx 2788 \text{ mm}^2 \quad (5)$$

2: Restrict to $x \geq 0$ half

The segment is symmetric about $x = 0$; the dead module only covers $x \geq 0$:

$$A_{\text{corner}} = \frac{A_{\text{seg}}}{2} \approx \frac{2788}{2} \approx 1394 \text{ mm}^2 \quad (6)$$

3: Total circle area

$$A_{\text{circle}} = \pi r^2 = \pi(90)^2 \approx 25447 \text{ mm}^2 \quad (7)$$

4: Fraction lost

$$f = \frac{A_{\text{corner}}}{A_{\text{circle}}} = \frac{1394}{25447} \approx 0.0548 \Rightarrow \boxed{5.5\%} \quad (8)$$

- 5.5% of the FV covered by the "dead" module.
- Assuming uniform distribution of tracks (we know its not): 5.5% of tracks will have -2 DoF.
- Likely to impact tracking performance (momentum, charge id).
- Effect to be quantified.

Tracker Meeting

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Figure: See Tracker Meeting Aug 25 2026

Track Quality

